Type MS · Area between a curve and the x-axis

Answer 1 of 7 · Question 10 part d · 4 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

Number		
10 (a)	$\frac{1}{2}+\sin 3x=0 \Rightarrow \sin 3x=-\frac{1}{2} \Rightarrow 3x=-\frac{\pi}{6},\frac{7\pi}{6},\frac{11\pi}{6} \Rightarrow x=\frac{7\pi}{18},\frac{11\pi}{18}$	M1
	Coordinates of M are $\left(\frac{7\pi}{18},\ 0\right)$*	A1
	Coordinates of N are $\left(\frac{11\pi}{18},\ 0\right)$	A1 [3]
	ALT	
	$\frac{1}{2}+\sin\left(3\times\frac{7\pi}{18}\right)=0$	[M1
	Coordinates of M are $\left(\frac{7\pi}{18},0\right)$ *	A1
	Coordinates of N are $\left(\frac{11\pi}{18},0\right)$	A1]
(b)	$\frac{dy}{dx}=3\cos 3x=0 \Rightarrow 3x=\frac{\pi}{2} \Rightarrow x=\frac{\pi}{6}$	M1A1
	$y=\frac{1}{2}+\sin 3\left(\frac{\pi}{6}\right)=1.5$	dM1A1 [4]
	$\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ 1.5\right)\right]$	
	ALT	
	Max of sine curve is 1 so that $\frac{1}{2}+1=\frac{3}{2} \Rightarrow y=\frac{3}{2}$	[M1A1
	$\sin 3\theta = 1 \ \Rightarrow\ 3\theta=\frac{\pi}{2} \Rightarrow\ \theta=\frac{\pi}{6}$	dM1A1]
	$\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ \frac{3}{2}\right)\right]$	
(c)	Uses the given x coordinate for point M	
	At point M $\quad\frac{dy}{dx}=3\cos\left(3\times\frac{7\pi}{18}\right)=-\frac{3\sqrt{3}}{2}$	B1
	$y-0=-\frac{3\sqrt{3}}{2}\left(x-\frac{7\pi}{18}\right) \Rightarrow 12y+18\sqrt{3}x-7\sqrt{3}\pi=0$ o.e.	M1A1A1 [4]
(d)	$A=\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2}+\sin 3x\right)dx+\left\lvert\left[\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2}+\sin 3x\right)dx\right]\right\rvert$	M1
	$A=\left[\frac{1}{2}x-\frac{\cos 3x}{3}\right]_0^{\frac{7\pi}{18}}+\left\lvert\left[\frac{1}{2}x-\frac{\cos 3x}{3}\right]_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\right\rvert$	M1
	$A=\left[\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{\cos 3\left(\frac{7\pi}{18}\right)}{3}\right)-\left(0-\frac{\cos 3\times 0}{3}\right)\right]$	M1

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	$+\left\lvert\left[\left(\frac{1}{2}\times\frac{11\pi}{18}-\frac{cos\,3\left(\frac{'11\pi'}{18}\right)}{3}\right)-\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\rvert$ $A=[(1.23287)+(0.22828)]=1.46115\ldots\approx 1.46$	A1 [4]
		Total 15 marks

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	A1	For correctly obtaining $x = \frac{\pi}{6}$
(c)	**B1**	For correct gradient at point M $\left(\frac{dy}{dx}\right) = -\frac{3\sqrt{3}}{2}$
	M1	For a fully correct method of finding the equation of the tangent to C at M $y - 0 = ' -\frac{3\sqrt{3}}{2} ' \left(x - \frac{7\pi}{18}\right)$ Allow use of their $-\frac{3\sqrt{3}}{2}$ obtained from substitution of $x = \frac{7\pi}{18}$ into their $\frac{dy}{dx}$.
	A1	For a correct equation of the tangent in any form: $y - 0 = -\frac{3\sqrt{3}}{2}\left(x - \frac{7\pi}{18}\right)$
	A1	For correct equation of the tangent in the required form: $12y + 18\sqrt{3}x - 7\sqrt{3}\pi = 0$ Accept integer multiples of this equation.
(d)	**M1**	For identifying the correct limits to find the area above the curve and the area below the curve: $\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) + \left|\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)\right|$ or $\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) - \int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)$ Allow use of their $\frac{11\pi}{18}$. Needs to correctly indicate dealing with areas above and below axis.
	M1	For an attempt to integrate $\frac{1}{2} + \sin 3x$ obtaining: $\frac{1}{2}x - k\cos 3x$ where $k \neq -3$ For this mark ignore incorrect / absent limits.
	M1	For substituting limits correctly into their integrated expression (must be a changed expression). $A = \left[\left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right) - \left(0 - \frac{cos\,3 \times 0}{3}\right)\right]$ $+ \left\|\left[\left(\frac{1}{2} \times \frac{11\pi}{18} - \frac{cos\,3\left(\frac{'11\pi'}{18}\right)}{3}\right) - \left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\|$ Allow use of their $\frac{11\pi}{18}$. Must show substitution or M0.
	A1	For correctly obtaining awrt 1.46

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Question number	Scheme	Marks

Type MS · Area between a curve and the x-axis

Answer 2 of 7 · Question 9 part c · 4 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

Number		
9(a)	$f'(x)=\lambda(x-2)(3x+1)$	M1
	$f(x)=\int\lambda(x-2)(3x+1)\,dx=\left[\int\lambda\left(3x^2-5x-2\right)\right]dx=\lambda\left(\frac{3x^3}{3}-\frac{5x^2}{2}-2x+k\right)$	M1A1
	At $(2,-9)$	
	$-9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=3$	M1
	$a=-5,b=-4,c=3\Rightarrow\left\{f(x)=2x^3-5x^2-4x+3\right\}$ *	A1*cso
		[5]
	ALT	
	$f(x)=2x^3+ax^2+bx+c$	
	$f'(x)=6x^2+2ax+b$	[M1
	Both $\begin{cases}0=6(2^2)+2(2)a+b\\0=6\times\left(-\frac{1}{3}\right)^2+2\times\left(-\frac{1}{3}\right)\times a+b\end{cases}$	M1A1
	$\begin{cases}72+12a+3b=0 \text{ and } 12-12a+18b=0\Rightarrow a=...,b=...\\\text{At } (2,9)\quad -9=2(2^3)-5(2^2)-4(2)+c\Rightarrow c=...\end{cases}$	M1
	$a=-5,b=-4,c=3\Rightarrow\left\{f(x)=2x^3-5x^2-4x+3\right\}$ *	A1*cso]
(b)(i)	$f(-1)=2(-1)^3-5(-1)^2-4(-1)+3=0$ oe	B1 [1]
(b)(ii)	$x+1\overline{)2x^3-5x^2-4x+3}$ $\quad 2x^2-7x+3$	M1
	$2x^2-7x+3=(2x-1)(x-3)\Rightarrow(x+1)(x-3)(2x-1)$	M1A1 [3]
(c)	$A=\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$	M1
	$A=\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{-1}^{\frac{1}{2}}+\left\lvert\left[\frac{1}{2}x^4-\frac{5}{3}x^3-2x^2+3x\right]_{\frac{1}{2}}^{3}\right\rvert$	M1
	$A=\left\{\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]-\left[\frac{1}{2}.(-1)^4-\frac{5}{3}.(-1)^3-2.(-1)^2+3.(-1)\right]\right\}$	M1
	$+\left\lvert\left\{\left[\frac{1}{2}.(3)^4-\frac{5}{3}.(3)^3-2.(3)^2+3(3)\right]-\left[\frac{1}{2}.\left(\frac{1}{2}\right)^4-\frac{5}{3}.\left(\frac{1}{2}\right)^3-2.\left(\frac{1}{2}\right)^2+3\left(\frac{1}{2}\right)\right]\right\}\right\rvert$	
	$A=\left\{\frac{117}{32}+\left\lvert-\frac{1375}{96}\right\rvert\right\}=\frac{863}{48}$	A1
		[4]
		Total 13 marks

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Answer 2 of 7 · Question 9 part c · 4 marks

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Part	Mark	Additional Guidance
(a)	M1	For forming an expression for $f'(x)$ using the two turning points given. Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	Attempts to integrate, see General Guidance for the definition of an attempt Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	A1	Correct integration including +c, Students may be working with $\lambda = 1$ at this stage and recognise the need to multiply through later.
	M1	$\lambda = 2$ is stated or implied, $\lambda k = c$ Substitution of $(2,-9)$ to their $f(x)$ (Does not have to be simplified) or substitution of $\left(-\frac{1}{3},\frac{100}{27}\right)$ (Does not have to be simplified)
	A1*cso	Obtains the **given** answer with no errors in the working
	ALT - starts with $f(x) = 2x^3 + ax^2 + bx + c$	
	M1	Attempts to differentiate, see General Guidance for the definition of an attempt
	M1	Equates differentiated expression to 0 and substitutes $x = 2$ **and** equates differentiated expression to 0 and substitutes $x = -\frac{1}{3}$
	A1	Two correct equations, simplified or unsimplified
	M1	Attempts to solve two **correct** equations simultaneously (including using calculators) to find a, b **and** substitution of their $a, b, (2,-9)$ to $f(x)$ to find c or substitution of their $a, b, \left(-\frac{1}{3},\frac{100}{27}\right)$ to $f(x)$ to find c
	A1*cso	Obtains the **given** answer with no errors in the working
(b)(i)	B1	Demonstrating that $x+1$ is a factor $f(-1) = 2(-1)^3 - 5(-1)^2 - 4(-1) + 3 = 0$ (factor theorem) If shown via polynomial division, the division must be fully correct and obtains 0 as the remainder.
(b)(ii)	M1	Dividing by $(x+1)$ to obtain a three-term quadratic quotient: $2x^2 - 7x + ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line : e.g. $(x+1)(x-3)(2x-1)$
	ALT I – equates coefficients	
	M1	For stating $2x^3 - 5x^2 - 4x + 3 = (x+1)(Ax^2 + Bx + C) \Rightarrow$ $2x^3 - 5x^2 - 4x + 3 = Ax^3 + (A+B)x^2 + (B+C)x + C \Rightarrow$ $A = 2, B = -7, C = ...$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$
	ALT II - inspection	
	M1	For finding the quadratic factor. Minimum required is $(x+1)(2x^2 - 7x + C)$
	M1	Attempt to factorise their 3TQ, see General Guidance. Or attempt to solve their 3TQ using the quadratic formula
	A1	Correct three linear factors written in one line: e.g. $(x+1)(x-3)(2x-1)$

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Answer 2 of 7 · Question 9 part c · 4 marks

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(c)	M1	For identifying the correct limits, uses a correct strategy to find the area above the x-axis and below the x-axis: $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx+\left\lvert\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx\right\rvert$ or $\int_{-1}^{\frac{1}{2}}(2x^3-5x^2-4x+3)\,dx-\int_{\frac{1}{2}}^{3}(2x^3-5x^2-4x+3)\,dx$ or implied by their working
	M1	For an attempt to integrate $2x^3-5x^2-4x+3$ See General Guidance for the definition of an attempt.
	M1	For clear substitution of **correct** limits into any **one** of their integrated expressions (must be a changed expression). Can be implied by **one** of the **correct exact** areas found from a **correctly** integrated expression
	A1	For correctly obtaining $\frac{863}{48}$ or $17\frac{47}{48}$

Type MS · Area between a curve and the x-axis

Answer 2 of 7 · Question 9 part c · 4 marks

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Question number	Scheme	Marks

Type MS · Area between a curve and the x-axis

Answer 3 of 7 · Question 11 part c · 3 marks

2024 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(\theta)=(2\cos\theta-\sin\theta)(2\sin\theta+\cos\theta)$ $=4\sin\theta\cos\theta+2\cos^2\theta-2\sin^2\theta-\sin\theta\cos\theta$ $=3\sin\theta\cos\theta+2(\cos^2\theta-\sin^2\theta)\Rightarrow \frac{3}{2}\sin 2\theta,+2\cos 2\theta$ $=\frac{3}{2}\sin 2\theta+2\cos 2\theta$ *	M1 M1A1 cso [3]
(b)	$\frac{3}{2}\sin 2\theta+2\cos 2\theta+2=0\Rightarrow 3\sin 2\theta+4\cos 2\theta+4=0$ $\Rightarrow 3\sin 2\theta+4(\cos^2\theta-\sin^2\theta)+4(\sin^2\theta+\cos^2\theta)=0$ $\Rightarrow 6\sin\theta\cos\theta+8\cos^2\theta=0$ $\Rightarrow \cos\theta(6\sin\theta+8\cos\theta)=0$ $\Rightarrow \cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$ * $\left[\Rightarrow \tan\theta=-\frac{4}{3}\right]$	M1M1 M1 M1 A1 cso [5]
(c)	$\text{Area}=\int_0^{\frac{\pi}{2}}\left(\frac{3}{2}\sin 2\theta+2\cos 2\theta+2\right)d\theta=\left[-\frac{3}{4}\cos 2\theta+\frac{2\sin 2\theta}{2}+2\theta\right]_0^{\frac{\pi}{2}}$ $=\left(-\frac{3}{4}\cos 2\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+2\left(\frac{\pi}{2}\right)\right)-\left(-\frac{3}{4}\cos 0+\frac{2\sin 0}{2}+2\times 0\right)$ $=\left(\frac{3}{4}+\pi\right)-\left(-\frac{3}{4}\right)=\frac{3}{2}+\pi$	M1 M1 A1 [3]
		Total 11 marks

Type MS · Area between a curve and the x-axis

Answer 4 of 7 · Question 3 part whole · 5 marks

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3	$9-x^2=0\Rightarrow x=\pm 3$	B1
	$A=\int_{'-3'}^{'3'} 9-x^2\ \mathrm{d}x=\left[9x-\frac{x^3}{3}\right]_{'-3'}^{'3'}$	M1A1
	$A=\left(9\times 3-\frac{3^3}{3}\right)-\left(9\times -3-\frac{[-3]^3}{3}\right)=36$	M1A1 [5]
	Total	**5 marks**

Mark	Notes
You may see the working without A = …. . Please accept for full marks.	
B1	Find the intersections of C with the x-axis. $9-x^2=0\Rightarrow x=\pm 3$
M1	Attempt to integrate the given expression only i.e., $9-x^2$ A squared expression integrated is M0. See General Guidance for the definition of an attempt. $(A)=\int_{'-3'}^{'3'} 9-x^2\ \mathrm{d}x=\left[9x-\frac{x^3}{3}\right]_{'-3'}^{'3'}$ Ignore limits for this mark, even if they are completely missing.
A1	For the correct integral $(A)=\left[9x-\frac{x^3}{3}\right]$ [ignore limits for this mark].
M1	For substituting **their** limits into **their** integrated expression provided it is changed from $9-x^2$ and is not a differentiated expression. $(A)=\left(9\times '3'-\frac{'3'^3}{3}\right)-\left(9\times '[-3]'-\frac{['-3']^3}{3}\right)=\ldots$ This mark can be implied by the correct final answer following correct integration. Allow also double 18 for this mark. If the final answer is incorrect and their integrated expression is incorrect, do not award this mark unless substitution is explicitly seen.
A1 [5]	For the correct answer only $(A)=36\ \ (\mathrm{cm}^2)$

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Answer 4 of 7 · Question 3 part whole · 5 marks

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Question	Scheme	Mark

Type MS · Area between a curve and the x-axis

Answer 5 of 7 · Question 10 part b · 8 marks

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Number		
10 (a)	$\sin(A+B)+\sin(A-B)=\sin A\cos B+\sin B\cos A+\sin A\cos B-\sin B\cos A$ $=2\sin A\cos B$	M1
	$\left[\sin(A+B)+\sin(A-B)=\sin 5x+\sin 3x\right]$ $Ax+Bx=5x$ $Ax-Bx=3x$	M1
	$\Rightarrow A=4x,\quad B=x$	
	$\Rightarrow \sin(4x+x)+\sin(4x-x)=\sin 5x+\sin 3x=2\sin 4x\cos x$	A1 cso [3]
	ALT	
	$\sin(A+B)-\sin(A-B)=\sin A\cos B+\sin B\cos A-(\sin A\cos B-\sin B\cos A)$ $=2\sin B\cos A$	[M1
	$\left[\sin(A+B)+\sin(A-B)=\sin(A+B)-\sin\left[-(A-B)\right]=\sin 5x+\sin 3x\right]$ $\Rightarrow A=x,\quad B=4x$	M1
	$\Rightarrow \sin(x+4x)-\sin(x-4x)=\sin 5x-\sin(-3x)=\sin 5x+\sin 3x=2\sin 4x\cos x$ $\left[-\sin(-3x)=+\sin 3x\right]$	A1cso]
(b)	**Intersections on x-axis** $6\sin x\cos x=0\Rightarrow \sin 4x=0,\quad \cos x=0$	M1
	from $\sin 4x=0\Rightarrow 4x=0,\pi,2\pi,3\pi,4\pi\rightarrow x=0,\frac{\pi}{4},\frac{\pi}{2},\frac{3\pi}{2},2\pi$ **and**	M1
	from $\cos x=0\Rightarrow x=0,\frac{\pi}{2},\frac{3\pi}{2}$ $\Rightarrow x=0,\frac{\pi}{4},\frac{\pi}{2}$	A1
	Integration $(3)\int(\sin 5x+\sin 3x)\,dx=(3)\left(-\frac{\cos 5x}{5}-\frac{\cos 3x}{3}\right)$	M1 A1 ft
	Limits $\int_0^{\frac{\pi}{4}}(\sin 5x+\sin 3x)\,dx\pm\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}(\sin 5x+\sin 3x)\,dx$	M1
	Substitution of values	

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	$3\left\{\left[\left(-\frac{\cos 5\left(\frac{\pi}{4}\right)}{5}-\frac{\cos 3\left(\frac{\pi}{4}\right)}{3}\right)-\left(-\frac{\cos 5(0)}{5}-\frac{\cos 3(0)}{3}\right)\right]-\left[\left(-\frac{\cos 5\left(\frac{\pi}{2}\right)}{5}-\frac{\cos 3\left(\frac{\pi}{2}\right)}{3}\right)-\left(-\frac{\cos 5\left(\frac{\pi}{4}\right)}{5}-\frac{\cos 3\left(\frac{\pi}{4}\right)}{3}\right)\right]\right\}$	ddM1
	$= 2.73137... + \lvert -1.313170... \rvert = 3.8267... \approx 3.86$	A1 cao [8]
		Total 11 marks

Part	Mark	Additional Guidance
(a)	M1	Use of both correct trig identities [either adding or subtracting], allow any two distinct letters or symbols and simplify to reach either $2\sin A\cos B$ or $2\sin B\cos A$ Allow use of $4x$ and x in place of A and B
	M1	Finds or deduces that $A = 4x, \quad B = x$ Or if they use the **ALT** scheme then $A = x, \quad B = 4x$ This may be implied by the next step or just written down.
	A1cso	For a final correct statement with no errors seen.
(b)	M1	(Placing the equation = 0) and arriving at one of $\sin 4x = 0$ ***or*** $\cos x = 0$
	M1	At least any **one** correct value for x from $\sin 4x = 0$ **and** at least any **one** correct value for x from $\cos x = 0$
	A1	Correctly identifies the required intersections of the curve with the x-axis either explicitly here or used in limits in following work, 0 need not be explicitly stated.
	Note: Award for these values seen without working, and allow working here in degrees.	
	M1	**Integration** An attempt at integration of both terms from their part (a) to an expression of the form $\pm\frac{\cos 5x}{5}\pm\frac{\cos 3x}{3}$ Ignore: • the presence or absence of the multiplier of 3 for this mark • any limits • incorrect notation
	A1	Fully correct integration. Again, ignore the presence/absence of the multiplier of 3. $(3)\left(-\frac{\cos 5x}{5}-\frac{\cos 3x}{3}\right)$
	M1	**Limits** Candidate clearly shows the integration of the curve needs to be done in two parts, where their limits are explicitly shown and come from the limits identified from solving the equation $6\sin x\cos x = 0$.

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		As a minimum we need to see two integrals of the function (this can either be written in the original form or the changed form). Their limits must be clearly shown for this mark (following through their solutions) and the upper limit on the first integral must be equal to the lower limit on the second. The candidate does not need to show that the modulus of the second integral must be taken for this mark. Ignore the presence/absence of the multiplier of 3.
	ddM1	Substitution of all limits –dependent on the previous two method marks, allow slips in simplification. The substitution must be correct. If you see any of the decimal answers shown in the working column or any of $\frac{8+4\sqrt{2}}{5}$ (1st integral) or $\frac{4\sqrt{2}}{5}$ (2nd integral) or $\frac{8+8\sqrt{2}}{5}$ (sum of integrals) award this mark. **NOTE: They must consider the modulus of the second integral, and the multiplier of 3 must also be taken into account for this mark.**
	A1cao	For awrt 3.86

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Question number	Scheme	Marks

Type MS · Area between a curve and the x-axis

Answer 6 of 7 · Question 5 part d · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 5 · find part d in this mark scheme extract

Question	Scheme	Marks
5(a)	$f(1)=2\times1^3+a\times1^2-14\times1+b=0 \quad \Rightarrow \quad a+b=12$ $f(4)=2\times4^3+a\times4^2-14\times4+b=39 \Rightarrow 16a+b=-33$ $\Rightarrow 15a=-45 \Rightarrow a=-3^*$ $\Rightarrow b=15$	M1 M1 M1A1 cso B1 [5]
	ALT – polynomial division	
	$(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **or** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $(2x^3+ax^2-14x+b)\div(x-1)=2x^2+(a+2)x-14+a+2$ and comparison of final step of division with b to obtain an equation **and** $(2x^3+ax^2-14x+b)\div(x-4)=2x^2+(a+8)x+(18+4a)$ r 39 and comparison of final step of division with obtaining remainder 39 to obtain an equation $b=12-a$ $b+4(18+4a)=39$ $a=-3^*$ $b=15$	[M1 M1 M1 A1cso B1]
(b)	$(x-1)\overline{)2x^3-3x^2-14x+15}$ $\quad 2x^2-x-15$ $f(x)=(x-1)(2x+5)(x-3)$	M1A1 dM1A1 [4]
(c)	$p=-\frac{5}{2} \quad q=1 \quad r=3$	B1ft B1ft B1ft [3]

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Answer 6 of 7 · Question 5 part d · 4 marks

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(d)	$A = \left\lvert \int_1^3 \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x \right\rvert + \int_{-\frac{5}{2}}^{1} \left(2x^3 - 3x^2 - 14x + 15\right) \mathrm{d}x$	M1
	$A = \left\lvert \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_1^3 \right\rvert + \left[\frac{2x^4}{4} - \frac{3x^3}{3} - \frac{14x^2}{2} + 15x\right]_{-\frac{5}{2}}^{1}$	M1*
	$A = \left\lvert \left[\left(\frac{3^4}{2} - 3^3 - 7\times 3^2 + 15\times 3\right) - \left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right)\right] \right\rvert$ $+ \left[\left(\frac{1^4}{2} - 1^3 - 7\times 1^2 + 15\times 1\right) - \left(\frac{\left(-\frac{5}{2}\right)^4}{2} - \left(-\frac{5}{2}\right)^3 - 7\times\left(-\frac{5}{2}\right)^2 + 15\times\left(-\frac{5}{2}\right)\right)\right]$	dM1
	$A = 12 + \frac{1715}{32} = \frac{2099}{32}$ or 65.59375	A1 [4]
		Total 16 marks

Type MS · Area between a curve and the x-axis

Answer 6 of 7 · Question 5 part d · 4 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 5 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	M1	For use of f (1) = 0 **or** use of f (4) = 39
	M1	For use of f (1) = 0 **and** use of f (4) = 39
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained. Must see an equation in just one variable.
	A1cso	For showing $a = -3$* Must see expression only in terms of a
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	ALT – polynomial division	
	M1	Correct method for forming one equation by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	Correct method for forming **both** equations by polynomial division together with comparison of the final step of the division with the required result (no remainder for division by $(x-1)$ and remainder 39 for division by $(x-4)$)
	M1	For solving the simultaneous equations in a and b Must see at least one line of working before a or b obtained.
	A1cso	For showing $a = -3$* Must see expression only in terms of a ie $ka = -3k$ before $a = -3$
	B1	For the correct value of $b = 15$ seen. Note: this is an A mark in epen.
	Assuming $a = -3$ and solving to find $b = 15$ scores maximum M1M0M0A0A1	
(b)	**Condone (b) completed in (c)**	
	M1	For carrying out polynomial division/comparing coefficients to find the quadratic factor. Must reach $2x^2 \pm \cdots$. Allow statement of quadratic factor if polynomial division by $(x-1)$ used in (a).
	A1	For the correct quadratic factor.
	dM1	For factorising their 3TQ See general guidance for what constitutes an attempt to factorise a 3TQ.
	A1	For the correct factorisation of f (x) on one line. Condone omission of f$(x) =$
	Note: complete factorisation seen with no method scores M0A0M0A0.	
(c)	B1ft	For one correct value for any of p, r or q FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For two correct values. FT their linear factors in (b). Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.
	B1ft	For all three correct values. FT their linear factors in (b), but **1 must be one of the values**. Values need not be identified as p, r, q, but if they are labelled these should be $p < q < r$.

Type MS · Area between a curve and the x-axis

Answer 6 of 7 · Question 5 part d · 4 marks

2023 JUN · Paper 01 (regional R) · extract 4 of 5 · find part d in this mark scheme extract

(d)	M1	For a correct statement for the area of the shaded regions with their limits correctly used. Minimum required is: $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x - \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x$ with $p < q < r$ shown as numerical limits or $\int_p^q (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x + \left\lvert \int_q^r (2x^3 - 3x^2 - 14x + 15)\mathrm{d}x \right\rvert$ with $p < q < r$ shown as numerical limits
	M1*	For an attempt to integrate their expression. Minimum required is: • two powers of terms to increase by 1 • no powers of terms to decrease Condone $+c$
	dM1	For substituting their limits correctly to evaluate the integrated expression. Must show the substitution if their integral or their limits are incorrect. Correct exact answer following correct integral with correct limits implies this mark. Condone $+c$ as long as this is cancelled out during the substitution.
	A1	For the correct area. Exact area required.

Type MS · Area between a curve and the x-axis

Answer 6 of 7 · Question 5 part d · 4 marks

2023 JUN · Paper 01 (regional R) · extract 5 of 5 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Area between a curve and the x-axis

Answer 7 of 7 · Question 7 part c · 4 marks

2025 NOV · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

7(a)	$\left[f\left(-\frac{1}{2}\right)=\right]2\times\left(-\frac{1}{2}\right)^3+A\times\left(-\frac{1}{2}\right)^2+B\times\left(-\frac{1}{2}\right)-20=0\left[\Rightarrow A-2B=81\right]$	M1
	$\left[f'(x)=\right]6x^2+2Ax+B$	M1
	$\left[f'(1)=\right]6\times1^2+2A\times1+B=-27\left[\Rightarrow 2A+B=-33\right]$	M1
	$A-2B=81$ oe	
	$2A+B=-33$ oe	dddM1
	$\Rightarrow A=3^*,\quad B=-39$	A1 cso [5]
(b)	$2x+1\overline{)2x^3+3x^2-39x+20}$ $\quad x^2+x-20$	M1
	$x^2+x-20=(x-4)(x+5)$	dM1
	$\Rightarrow\left[x=\right]-\frac{1}{2},4,-5$	A1 [3]
(c)	$\left[A=\right]\int_{"-5"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)dx+\left\lvert\int_{"-\frac{1}{2}"}^{"4"}\left(2x^3+3x^2-39x-20\right)dx\right\rvert$	M1
	oe eg $\int_{"-5"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)dx+\int_{"4"}^{"-\frac{1}{2}"}\left(2x^3+3x^2-39x-20\right)dx$	
	$\left[A=\right]\left[\frac{2x^4}{4}+\frac{3x^3}{3}-\frac{39x^2}{2}-20x\right]_{[-5]}^{\left[-\frac{1}{2}\right]}+\left[\frac{2x^4}{4}+\frac{3x^3}{3}-\frac{39x^2}{2}-20x\right]_{\left[-\frac{1}{2}\right]}^{[4]}$	M1
	$\left[A=\right]\left\{\left[\frac{2\left(-\frac{1}{2}\right)^4}{4}+\frac{3\left(-\frac{1}{2}\right)^3}{3}-\frac{39\left(-\frac{1}{2}\right)^2}{2}-20\left(-\frac{1}{2}\right)\right]-\left[\frac{2(-5)^4}{4}+\frac{3(-5)^3}{3}-\frac{39(-5)^2}{2}-20(-5)\right]\right\}$	
	$\left[+\right]\left\lvert\left\{\left[\frac{2(4)^4}{4}+\frac{3(4)^3}{3}-\frac{39(4)^2}{2}-20(4)\right]-\left[\frac{2\left(-\frac{1}{2}\right)^4}{4}+\frac{3\left(-\frac{1}{2}\right)^3}{3}-\frac{39\left(-\frac{1}{2}\right)^2}{2}-20\left(-\frac{1}{2}\right)\right]\right\}\right\rvert$ oe	M1
	$\left[A=\right]\frac{6561}{16}$ or 410.0625 oe	A1 [4]
		Total 12 marks

Type MS · Area between a curve and the x-axis

Answer 7 of 7 · Question 7 part c · 4 marks

2025 NOV · Paper 02 · extract 2 of 4 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	Substitutes $-\frac{1}{2}$ into f(x) and sets = 0 Accept as a minimum $2\times\left(-\frac{1}{8}\right)+A\times\frac{1}{4}+B\times\left(-\frac{1}{2}\right)-20=0$ or $-\frac{1}{4}+A\times\frac{1}{4}+B\times\left(-\frac{1}{2}\right)-20=0$
	M1	For correctly differentiating f(x)
	M1	Substitutes 1 into their derivative, and sets = – 27 Accept $2A+B=-33$ with no errors. This isn't a dependent mark, but candidates must be substituting into a changed expression with at least 2 terms in x
	dddM1	Solves their linear simultaneous equations by any method, allow one error in solving. Minimal working here is fine, but there must be **some** working. Candidates who show no working to solve the simultaneous equations will not gain this mark (and therefore not the next mark) As this is a 'show' question, correct answers cannot imply this method mark. Dependent on previous 3 method marks.
	A1*cso	For the correct value of A [which is given in the question] and the correct value of B. Minimum steps as shown/stated, no errors or omissions.
(b)	M1	Award the mark if the candidate is seen to be attempting any method, such as algebraic division or equating coefficients (note this will often have minimal working) and attains: $x^2\pm Cx-20$ or $x^2+x\pm D$ $C,D\neq 0$ It is not sufficient to show simply a product of linear factors.
	dM1	For correctly factorising their quadratic factor or correctly using any method to solve their quadratic factor = 0 Dependent on previous method mark.
	A1	For all three roots. Candidates are told to use algebra, so correct answers alone cannot imply M marks for this question. This means correct answers alone will be 0 marks overall.
The majority of candidates are completing this question, using the factor of $(2x + 1)$ given. Occasionally, we are seeing a re-start using a different factor – probably deduced from a calculator, but if sufficient algebra is shown, we will award the marks. For $(x + 5)$ need to see $2x^2-7x-4$ with $2x^2$ correct and either the – 7 or – 4 correct For $(x – 4)$ need to see $2x^2+11x+5$ with $2x^2$ correct and either the +11 or +5 correct for the first M mark – the next 2 marks then follow.		

Type MS · Area between a curve and the x-axis

Answer 7 of 7 · Question 7 part c · 4 marks

2025 NOV · Paper 02 · extract 3 of 4 · find part c in this mark scheme extract

(c)	M1	For a correct statement of the required area. Allow use of their limits from part (b), even if working wasn't shown. However, no limit may be 0 This mark may be implicit in further working and may even be awarded at the stage when a negative numerical area has been made positive.
	M1	For a minimally acceptable attempt at integration of an expression of the form $2x^3+3x^2+Px-20$ $P\neq 0$. The modulus sign and the addition between the 2 calculations do not need to be seen for this mark. See general guidance, additionally, no power of x to decrease. Limits do no need to be present for this mark.
	M1	For a full and correct substitution of each of their limits (their roots from part (b), no limit 0) into any changed expression with a minimum of 2 terms. The modulus sign and the addition between the 2 calculations do not need to be seen for this mark. A correct final answer, if the integration is shown, may imply this mark.
	A1	For the correct exact area. The correct final answer may imply the first and third method marks, but may not imply the second method mark (ie must show integration). If no integration is shown, with a correct answer, the only possible marks are M1 M0 M1 A0 (if it's clear somewhere in working that the candidate has split the area). or M0 M0 M1 A0 (if the answer is correct with no reference to splitting the area).
If candidates work with $\int_{"-5"}^{"4"}\left(2x^3+3x^2-39x-20\right)\mathrm{d}x$ they **may** attain up to M0 M1 M1 A0		

Type MS · Area between a curve and the x-axis

Answer 7 of 7 · Question 7 part c · 4 marks

2025 NOV · Paper 02 · extract 4 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks